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1 &

1.1

$m(t)$ (baseband) (1)

$s(t)$ (2)

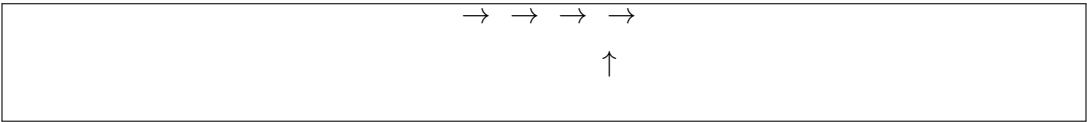
$r(t)$ (3)

$\hat{m}(t)$ (4)

f_c (carrier frequency) (5)

$m(t) \rightarrow \rightarrow s(t) \rightarrow \rightarrow r(t) \rightarrow \rightarrow \hat{m}(t)$ (6)

$s(t) \rightarrow + \rightarrow r(t)$ (7)



$: \hat{m}(t) = m(t)$ (8)

.

1.2 &

$: m(t) \xrightarrow{\text{modulation}} f_c$ (9)

Baseband $\rightarrow$ Passband (10)

$: \text{Passband} \rightarrow \text{Baseband}$ (11)

$r(t) \rightarrow \hat{m}(t)$ (12)

1.3 6

1.

2.

3.

4.

5.

6.

1.4

$\lambda = \frac{c}{f}, \quad c = 3 \times 10^8 \text{ m/s}$ (13)

1.5

VLF	3 – 30 kHz
LF	30 – 300 kHz
MF	300 kHz – 3 MHz
HF	3 – 30 MHz
VHF	30 – 300 MHz
UHF	300 MHz – 3 GHz
SHF	3 – 30 GHz
EHF	30 – 300 GHz

1.6

Simplex, Half-Duplex, Full-Duplex.

2 Session 3 : (theory1)

2.1

$$x \triangleq \{x[n]\}, \quad -\infty < n < +\infty \quad (14)$$

2.2

$$x(t) = x(t + kT), \quad \forall t, k \in \mathbb{N} \quad (15)$$

2.3

$$x[n] = x[n + kN], \quad \forall n, k \in \mathbb{N} \quad (16)$$

2.4

$$s_1(t) = s_1(t + k_1 T_1), \quad s_2(t) = s_2(t + k_2 T_2) \quad (17)$$

$$k_1 T_1 = k_2 T_2 = T, \quad \frac{T_1}{T_2} = \frac{k_2}{k_1} \quad (18)$$

$$T_1/T_2 \quad .$$

2.5 $\sin(\omega n)$

$$x[n] = \sin(\omega n) \quad (19)$$

$$\omega N = 2\pi m, \quad m \in \mathbb{Z}^+ \quad (20)$$

$$\frac{\omega}{2\pi} = \frac{m}{N} \quad (21)$$

$$2\pi/\omega \quad .$$

2.6

$$x(t) = Ae^{j2\pi f_c t} \quad (22)$$

2.7 Euler

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (23)$$

$$x(t) = A \cos(2\pi f_c t) + jA \sin(2\pi f_c t) \quad (24)$$

2.8 In-Phase & Quadrature

$$x_I(t) = A \cos(2\pi f_c t) \quad (25)$$

$$x_Q(t) = A \sin(2\pi f_c t) \quad (26)$$

2.9

$$|x(t)| = \sqrt{x_I^2(t) + x_Q^2(t)} \quad (27)$$

2.10

$$\theta(t) = \tan^{-1} \left(\frac{x_I(t)}{x_Q(t)} \right) \quad (28)$$

2.11 IQ

$$x(t) = A(t) \cos(2\pi f_c t + \theta(t)) \quad (29)$$

2.12

$$\cos(a + b) = \cos(a) \cos(b) - \sin(a) \sin(b) \quad (30)$$

2.13 IQ decomposition

$$x(t) = A(t) \cos \theta(t) \cos(2\pi f_c t) - A(t) \sin \theta(t) \sin(2\pi f_c t) \quad (31)$$

2.14 &

$$x(-t) = x(t) \quad () \quad (32)$$

$$x(-t) = -x(t) \quad () \quad (33)$$

$$x(t) = x_e(t) + x_o(t) \quad (34)$$

2.15 &

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)] \quad (35)$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)] \quad (36)$$

2.16

$$x(t) = 0, \quad t < 0 \quad (37)$$

2.17

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad (38)$$

$$E_x = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt \quad (39)$$

2.18

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \quad (40)$$

2.19

$$P_x = \frac{1}{2T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt \quad (41)$$

2.20 &

$$0 < E_x < \infty \quad () \quad (42)$$

$$0 < P_x < \infty \quad () \quad (43)$$

2.21

$$x(t) = A \cos(2\pi f_c t) \quad (44)$$

2.22

$$T_c = \frac{1}{f_c} \quad (45)$$

2.23 $\cos^2$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2} \quad (46)$$

$$\cos^2(2\pi f_c t) = \frac{1 + \cos(4\pi f_c t)}{2} \quad (47)$$

2.24 &

$$P_x = \frac{A^2}{2} \quad (48)$$

$$E_x = \infty \quad (49)$$

2.25 DC

$$P_x^{\text{DC}} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt \quad (50)$$

$$P_x^{\text{DC}} = \frac{1}{2T_0} \int_{-T_0/2}^{T_0/2} x(t) dt \quad () \quad (51)$$

2.26 RMS

$$P_x^{\text{RMS}} = \sqrt{P_x} \quad (52)$$

2.27

$$x(t) = Ae^{-\alpha t} u(t) \quad (53)$$

2.28 Dirac delta

$$p(t) = \frac{1}{\epsilon} \text{rect}\left(\frac{t}{\epsilon}\right) \quad (54)$$

$$\delta(t) = \lim_{\epsilon \rightarrow 0} p(t) \quad (55)$$

2.29 $\delta(t)$

$$\delta(t) = 0, \quad t \neq 0 \quad (56)$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (57)$$

$$\delta(-t) = \delta(t) \quad (58)$$

$$\delta(at) = \frac{1}{|a|} \delta(t) \quad (59)$$

2.30 Sifting property

$$\int_{-\infty}^{\infty} f(t) \delta(t - \tau) dt = f(\tau) \quad (60)$$

2.31 Fourier δ

$$\mathcal{F}\{\delta(t)\} = 1 \quad (61)$$

$$\mathcal{F}\{1\} = \delta(f) \quad (62)$$

$$\mathcal{F}\{e^{j2\pi f_0 t}\} = \delta(f - f_0) \quad (63)$$

3 Session 4 & (theory2)

3.1

$$h(t) = \mathcal{S}\{\delta(t)\} \quad (64)$$

3.2

$$y(t) = x(t) * h(t) \quad (65)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \quad (66)$$

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \quad (67)$$

3.3

$$\delta(t) * h(t) = h(t) \quad (68)$$

$$h(t) * \delta(t - t_0) = h(t - t_0) \quad (69)$$

$$x(t) * \delta(t - t_0) = x(t - t_0) \quad (70)$$

$$[h_1(t) + h_2(t)] * x(t) = h_1(t) * x(t) + h_2(t) * x(t) \quad (71)$$

$$(z(t) * x(t)) * y(t) = z(t) * (x(t) * y(t)) \quad (72)$$

$$x(t) * y(t) = y(t) * x(t) \quad (73)$$

3.4 4

$$x_1(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \end{cases} \quad (74)$$

$$x_2(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ -1, & 1 \leq t \leq 2 \\ 0, & \end{cases} \quad (75)$$

$$y(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) d\tau = \int_{t-1}^t x_2(\kappa) d\kappa \quad (76)$$

$$y(t) = \begin{cases} t, & t \in [0, 1] \\ 3 - 2t, & t \in [1, 2] \\ t - 3, & t \in [2, 3] \\ 0, & t \in (-\infty, 0] \cup [3, \infty) \end{cases} \quad (77)$$

3.5 Euler

$$x(t) = A \cos(2\pi f_0 t + \varphi), \quad A > 0, \varphi \in (-\pi, \pi] \quad (78)$$

$$x(t) = \frac{A}{2} e^{j\varphi} e^{j2\pi f_0 t} + \frac{A}{2} e^{-j\varphi} e^{-j2\pi f_0 t} \quad (79)$$

3.6

$$x(t) = A e^{j(2\pi f_0 t + \varphi)} \quad (80)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h(\tau) A e^{j2\pi f_0(t-\tau) + j\varphi} d\tau \quad (81)$$

$$H(f_0) = \int_{-\infty}^{\infty} h(t) e^{-j2\pi f_0 t} dt \quad (82)$$

$$y(t) = H(f_0) x(t) \quad (83)$$

3.7

$$x(t) = \sum_k A_k \cos(2\pi f_k t + \varphi_k) \quad (84)$$

$$x(t) = \sum_k \left[\frac{A_k}{2} e^{j\varphi_k} e^{j2\pi f_k t} + \frac{A_k}{2} e^{-j\varphi_k} e^{-j2\pi f_k t} \right] \quad (85)$$

3.8

$$\langle \mathbf{x}, \mathbf{y} \rangle = |\mathbf{x}| |\mathbf{y}| \cos \theta \quad (86)$$

$$\langle x(t), y(t) \rangle = \int_a^b x(t) y^*(t) dt = 0 \quad (87)$$

3.9

$$\left\langle e^{jk\omega_0 t}, e^{jm\omega_0 t} \right\rangle = \int_0^T e^{jk\omega_0 t} e^{-jm\omega_0 t} dt = \int_0^T e^{j(k-m)\omega_0 t} dt \quad (88)$$

$$= \frac{1}{j(k-m)\omega_0} \left[e^{j(k-m)\omega_0 T} - 1 \right] = 0, \quad k \neq m \quad (89)$$

$$e^{j(k-m)\omega_0 T} = e^{j(k-m)2\pi} = \cos((k-m)2\pi) - j \sin((k-m)2\pi) = 1, \quad k, m \in \mathbb{Z} \quad (90)$$

$$\left\langle e^{jk\omega_0 t}, e^{jm\omega_0 t} \right\rangle = T, \quad k = m \quad (91)$$

$$\left\langle e^{jk\omega_0 t}, e^{jm\omega_0 t} \right\rangle = T \delta(k-m) \quad (92)$$

3.10

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad (93)$$

$$a_n = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jn\omega_0 t} dt \quad (94)$$

3.11 Fourier

$$: \quad x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad (95)$$

$$: \quad a_k = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jk\omega_0 t} dt \quad (96)$$

$$a_k = |a_k| e^{j \arg(a_k)} \quad (97)$$

4 Session 5&6 Fourier, , Parseval,

4.1 Fourier ()

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad (98)$$

$$a_k = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jk\omega_0 t} dt \quad (99)$$

$$a_k = |a_k| e^{j \arg(a_k)} \quad (100)$$

4.2 Euler &

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (101)$$

$$A \cos(2\pi f_0 t + \phi) = \frac{A}{2} e^{j\phi} e^{j2\pi f_0 t} + \frac{A}{2} e^{-j\phi} e^{-j2\pi f_0 t} \quad (102)$$

4.3

$$H(f) = \int_{-\infty}^{\infty} h(t) e^{-j2\pi f t} dt \quad (103)$$

$$y(t) = H(f_0) x(t) \quad (104)$$

4.4 Fourier

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt \quad (105)$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df \quad (106)$$

4.5 &

$$X(f) = X_R(f) + jX_I(f) = \Re\{X(f)\} + j\Im\{X(f)\} \quad (107)$$

$$X(f) = \int x(t) \cos(2\pi ft) dt - j \int x(t) \sin(2\pi ft) dt \quad (108)$$

4.6 &

$$|X(f)| = \sqrt{X_R^2(f) + X_I^2(f)} \quad (109)$$

$$\phi(f) = \tan^{-1} \left(\frac{X_I(f)}{X_R(f)} \right) \quad (110)$$

$$X(f) = |X(f)| e^{j\phi(f)} \quad (111)$$

4.7

$$x(t) = A \operatorname{rect} \left(\frac{t}{T} \right) \quad (112)$$

$$X(f) = AT \operatorname{sinc}(fT) \quad (113)$$

$$: f = \pm \frac{1}{T}, \pm \frac{2}{T}, \pm \frac{3}{T}, \dots \quad (114)$$

4.8 sinc

$$\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x} \quad (115)$$

4.9 Fourier

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty \quad (116)$$

4.10 Fourier

$$x(at) \longleftrightarrow \frac{1}{|a|} X \left(\frac{f}{a} \right) \quad (117)$$

$$x(t - t_0) \longleftrightarrow e^{-j2\pi ft_0} X(f) \quad (118)$$

$$e^{j2\pi f_c t} x(t) \longleftrightarrow X(f - f_c) \quad (119)$$

$$a_1 x_1(t) + a_2 x_2(t) \longleftrightarrow a_1 X_1(f) + a_2 X_2(f) \quad (120)$$

$$x_1(t) x_2(t) \longleftrightarrow X_1(f) * X_2(f) \quad (121)$$

$$x_1(t) * x_2(t) \longleftrightarrow X_1(f) X_2(f) \quad (122)$$

$$\frac{d^k x(t)}{dt^k} \longleftrightarrow (j2\pi f)^k X(f) \quad (123)$$

4.11

$$y(t) = h(t) * x(t) \quad (124)$$

$$Y(f) = H(f) X(f) \quad (125)$$

4.12

$$e^{j2\pi f_c t} x(t) \longleftrightarrow X(f - f_c) \quad (126)$$

$$y(t) = Ax(t) \cos(2\pi f_c t) \quad (127)$$

$$Y(f) = \frac{A}{2} [X(f - f_c) + X(f + f_c)] \quad (128)$$

4.13 Fourier

$$\delta(t) \longleftrightarrow 1 \quad (129)$$

$$1 \longleftrightarrow \delta(f) \quad (130)$$

$$u(t) \longleftrightarrow \frac{\delta(f)}{2} + \frac{1}{j2\pi f} \quad (131)$$

$$\text{sgn}(t) \longleftrightarrow \frac{1}{j\pi f} \quad (132)$$

$$\cos(2\pi f_0 t) \longleftrightarrow \frac{1}{2} [\delta(f - f_0) + \delta(f + f_0)] \quad (133)$$

$$\sin(2\pi f_0 t) \longleftrightarrow \frac{1}{2j} [\delta(f - f_0) - \delta(f + f_0)] \quad (134)$$

$$\text{rect}(t) \longleftrightarrow \text{sinc}(f) \quad (135)$$

$$\Lambda(t) \longleftrightarrow \text{sinc}^2(f) \quad (136)$$

4.14 Parseval

$$P_x = \sum_{k=-\infty}^{\infty} |a_k|^2 \quad (137)$$

$$E_x = \int_{-\infty}^{\infty} |X(f)|^2 df = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad (138)$$

4.15

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t) y^*(t - \tau) dt \quad (139)$$

$$R_{yx}(\tau) = \int_{-\infty}^{\infty} y(t) x^*(t - \tau) dt \quad (140)$$

$$R_{xy}(\tau) = x(t) * y^*(-t) \quad (141)$$

4.16

$$R_{xy}(\tau) = R_{yx}^*(-\tau) \quad (142)$$

$$R_{xy}(0) = 0 \quad (143)$$

$$R_{xy}(\tau + kT_0) = R_{xy}(\tau) \quad (144)$$

4.17

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t) x^*(t - \tau) dt = \int_{-\infty}^{\infty} x(t + \tau) x^*(t) dt \quad (145)$$

$$R_x(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t - \tau) x^*(t) dt \quad (146)$$

4.18

$$R_x(-\tau) = R_x^*(\tau) \quad (147)$$

$$E_x = R_x(0) \quad (148)$$

$$|R_x(\tau)| \leq R_x(0) \quad (149)$$

$$R_x(0) = P_x \quad (150)$$

4.19 Wiener–Khinchin

$$\mathcal{F}\{R_x(\tau)\} = S_x(f) \quad (151)$$

$$\mathcal{F}\{R_x(\tau)\} = |X(f)|^2 \quad (152)$$

5 Session 7&8 Parseval, Hilbert, , IQ

5.1 Parseval (&)

$$P_x = \sum_{k=-\infty}^{\infty} |X[k]|^2 \quad (153)$$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df \quad (154)$$

5.2 &

$$0 < E_x < \infty \quad (155)$$

$$0 < P_x < \infty \quad (156)$$

5.3

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t) y^*(t - \tau) dt \quad (157)$$

$$R_{yx}(\tau) = \int_{-\infty}^{\infty} y(t) x^*(t - \tau) dt \quad (158)$$

5.4

$$R_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) y^*(t - \tau) dt \quad (159)$$

$$R_{yx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} y(t) x^*(t - \tau) dt \quad (160)$$

5.5

$$R_{xy}(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) y^*(t - \tau) dt \quad (161)$$

$$R_{yx}(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} y(t) x^*(t - \tau) dt \quad (162)$$

5.6 —

$$R_{xy}(\tau) = x(t) * y^*(-t) \quad (163)$$

5.7

$$R_{xy}(\tau) \neq R_{yx}(\tau) \quad (164)$$

$$R_{xy}(\tau) = R_{yx}^*(-\tau) \quad (165)$$

$$\int_{-\infty}^{\infty} x(t) y^*(t) dt = 0 \Rightarrow R_{xy}(0) = 0 \quad (166)$$

$$R_{xy}(\tau + kT_0) = R_{xy}(\tau) \quad (167)$$

5.8

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t) x^*(t - \tau) dt = \int_{-\infty}^{\infty} x(t + \tau) x^*(t) dt \quad (168)$$

$$R_x(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t - \tau) x^*(t) dt \quad (169)$$

5.9

$$R_x(-\tau) = R_x^*(\tau) \quad (170)$$

$$E_x = R_x(0) = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad (171)$$

$$|R_x(\tau)| \leq R_x(0) = E_x \quad (172)$$

$$R_x(0) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt = P_x \quad (173)$$

5.10 Wiener–Khinchin &

$$P_x = \int_{-\infty}^{\infty} S_x(f) df \quad (174)$$

$$\mathcal{F}\{R_x(\tau)\} = S_x(f) \quad (175)$$

$$\mathcal{F}\{R_x(\tau)\} = |X(f)|^2 \quad (176)$$

5.11 Hilbert

$$h(t) = \frac{1}{\pi t} \quad (177)$$

$$\mathcal{H}\{x(t)\} = x(t) * h(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau \quad (178)$$

$$\mathcal{F}\{\mathcal{H}[x(t)]\} = -j \operatorname{sgn}(f) X(f) \quad (179)$$

5.12 Hilbert

$$\mathcal{H}\{x_1 * x_2\} = \mathcal{H}\{x_1\} * x_2 = x_1 * \mathcal{H}\{x_2\} \quad (180)$$

$$\mathcal{H}\{\mathcal{H}[x(t)]\} = -x(t) \quad (181)$$

$$\mathcal{H}\{x(at)\} = \text{sgn}(a) \mathcal{H}\{x(t)\} \quad (182)$$

5.13 &

$$X(f) = 0, \quad |f| \geq W \quad () \quad (183)$$

$$X(f) = 0, \quad |f - f_c| \geq W \quad () \quad (184)$$

5.14

$$x(t) = Am(t) \cos(2\pi f_c t) \quad (185)$$

$$y(t) = Ax(t) \cos(2\pi f_c t) \quad (186)$$

$$Y(f) = \frac{A}{2} [X(f - f_c) + X(f + f_c)] \quad (187)$$

5.15

$$x_p(t) = x(t) + j \hat{x}(t), \quad \hat{x}(t) = \mathcal{H}\{x(t)\} \quad (188)$$

5.16

$$x(t) = \text{Re}\{g(t) e^{j2\pi f_c t}\} \quad (189)$$

$$g(t) = x_p(t) e^{-j2\pi f_c t} \quad (190)$$

$$x_p(t) = g(t) e^{j2\pi f_c t} \quad (191)$$

5.17 IQ

$$g(t) = x_I(t) + jx_Q(t) \quad (192)$$

$$x(t) = x_I(t) \cos(2\pi f_c t) - x_Q(t) \sin(2\pi f_c t) \quad (193)$$

5.18 &

$$V(t) = \sqrt{x_I^2(t) + x_Q^2(t)} \quad (194)$$

$$\theta(t) = \tan^{-1} \left(\frac{x_I(t)}{x_Q(t)} \right) \quad (195)$$

$$g(t) = V(t) e^{j\theta(t)} \quad (196)$$

$$x_I(t) = V(t) \cos \theta(t), \quad x_Q(t) = V(t) \sin \theta(t) \quad (197)$$

$$x(t) = V(t) \cos(2\pi f_c t + \theta(t)) \quad (198)$$

5.19 AM

$$g(t) = V(t) = A_c + m(t) \quad (199)$$

$$x_I(t) = A_c + m(t), \quad x_Q(t) = 0 \quad (200)$$

5.20

$$x(t) = \text{Re}\{g(t) e^{j2\pi f_c t}\} \Rightarrow X(f) = \frac{1}{2}[G(f - f_c) + G^*(-f - f_c)] \quad (201)$$

5.21 ()

$$S_x(f) = \frac{1}{4}[S_g(f - f_c) + S_g(-f - f_c)] \quad (202)$$

5.22 dB

$$\text{Gain(dB)} = 20 \log_{10} |H(f)| \quad (203)$$

6 Session 9 & WSS

6.1

$$m_X(t) = \mathbb{E}[X(t)] = \int_{-\infty}^{\infty} a f_{X(t)}(a) da \quad () \quad (204)$$

$$m_X(t) = \mathbb{E}[X(t)] = \sum_i a_i P[X(t) = a_i] \quad () \quad (205)$$

6.2 ()

$$R_X(t_i, t_j) = \mathbb{E}[X(t_i)X(t_j)] \quad (206)$$

$$R_X(t_i, t_j) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} ab f_{X(t_i), X(t_j)}(a, b) da db \quad (207)$$

$$R_X(t_i, t_j) = R_X(t_j, t_i) \quad (208)$$

6.3

$$C_X(t_i, t_j) = \mathbb{E}[(X(t_i) - m_X(t_i))(X(t_j) - m_X(t_j))] \quad (209)$$

$$C_X(t_i, t_j) = R_X(t_i, t_j) - m_X(t_i)m_X(t_j) \quad (210)$$

6.4 &

$$R_{XY}(t_1, t_2) = \mathbb{E}[X(t_1)Y(t_2)] \quad (211)$$

$$R_{YX}(t_1, t_2) = \mathbb{E}[Y(t_1)X(t_2)] \quad (212)$$

$$C_{XY}(t_1, t_2) = R_{XY}(t_1, t_2) - m_X(t_1)m_Y(t_2) \quad (213)$$

6.5 &

$$R_{XY}(t_1, t_2) = 0, \quad \forall t_1, t_2 \quad () \quad (214)$$

$$C_{XY}(t_1, t_2) = 0, \quad \forall t_1, t_2 \quad () \quad (215)$$

6.6 (SSS)

$$f_{X(t_1), X(t_2), \dots}(a_1, a_2, \dots) = f_{X(t_1+\tau), X(t_2+\tau), \dots}(a_1, a_2, \dots), \quad \forall \tau \quad (216)$$

6.7 WSS (Wide Sense Stationary)

$$m_X(t) = m_X = \quad (217)$$

$$R_X(t_i, t_j) = R_X(t_i - t_j) = R_X(\tau) \quad (218)$$

6.8 WSS

$$C_X(\tau) = R_X(\tau) - m_X^2 \quad (219)$$

6.9 WSS

$$X(t) = m_X + N(t), \quad \mathbb{E}[N(t)] = 0 \quad (220)$$

$$R_X(\tau) = m_X^2 + R_N(\tau) \quad (221)$$

6.10

$$\mathbb{E}[x_i(t)] = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x_i(t) dt \quad (222)$$

6.11

$$R_{x_i}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x_i(t) x_i(t + \tau) dt \quad (223)$$

6.12

$$\mathbb{E}[x_i(t)] = \mathbb{E}[X(t)] \Rightarrow \mathbb{E}[X(t)] = m_X \quad (224)$$

$$R_{x_i}(\tau) = R_X(\tau) \quad (225)$$

6.13 Wiener–Khinchin (PSD)

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau \quad (226)$$

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df \quad (227)$$

6.14

$$P_X = R_X(0) \quad (228)$$

$$P_X = \int_{-\infty}^{\infty} S_X(f) df \quad (229)$$

$$R_X(0) = \mathbb{E}[X^2(t)] \quad (\text{WSS}) \quad (230)$$

6.15 & WSS

$$Y(t) = X(t) * h(t) = \int_{-\infty}^{\infty} X(\tau) h(t - \tau) d\tau \quad (231)$$

$$R_Y(\tau) = R_X(\tau) * h(\tau) * h(-\tau) \quad (232)$$

$$S_Y(f) = S_X(f) |H(f)|^2 \quad (233)$$

$$P_Y = R_Y(0) = \int_{-\infty}^{\infty} S_X(f) |H(f)|^2 df \quad (234)$$

6.16 PSD

$$\mathcal{F}\{R_x(\tau)\} = |X(f)|^2 \quad (235)$$

7 Session 10

7.1 Gaussian PDF

$$f_N(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} \quad (236)$$

7.2

$$\sigma^2 = \mathbb{E}[N^2(t)] = 4kTRW \quad (237)$$

$$k = 1,38 \times 10^{-23} \text{ J/K}, \quad T \text{ Kelvin}, \quad R \text{ } \Omega, \quad W \text{ Hz} \quad (238)$$

7.3

$$\sigma^2 = \mathbb{E}[N^2(t)] = 4k(T_e + T_i)RW \quad (239)$$

7.4

$$P_N = \frac{\mathbb{E}[N^2(t)]}{4R} = kTW \quad (240)$$

7.5 PSD

$$S_N(f) = \frac{kT}{2} \text{ W/Hz} \quad (241)$$

7.6

$$S_N(f) = \frac{N_0}{2} \quad (242)$$

$$N_0 = kT \quad (243)$$

$$R_N(\tau) = \mathcal{F}^{-1}\{S_N(f)\} = \frac{N_0}{2} \delta(\tau) \quad (244)$$

7.7 8 +

$$S_N(f) = \frac{N_0}{2} \quad (245)$$

$$H(f) = 1, \quad f_c - \frac{W}{2} \leq |f| \leq f_c + \frac{W}{2} \quad (246)$$

$$S_Y(f) = S_N(f) |H(f)|^2 \quad (247)$$

$$R_Y(\tau) = N_0 W \cos(2\pi f_c \tau) \text{ sinc}(W\tau) \quad (248)$$

$$P_Y = R_Y(0) = N_0 W \quad (249)$$

8 15

1. $m_X(t) = \mathbb{E}[X(t)]$
2. $R_X(t_1, t_2) = \mathbb{E}[X(t_1)X(t_2)]$
3. $C_X = R_X - m_X m_X$
4. $R_{XY} = \mathbb{E}[XY]$
5. WSS: $m_X = \text{const}$, $R_X(t_1, t_2) = R_X(\tau)$
6. $C_X(\tau) = R_X(\tau) - m_X^2$
7. $: \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x_i(t) dt$
8. $: \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x_i(t)x_i(t + \tau) dt$
9. $S_X(f) = \mathcal{F}\{R_X(\tau)\}$
10. $R_X(\tau) = \mathcal{F}^{-1}\{S_X(f)\}$
11. $P_X = R_X(0) = \int S_X(f) df$
12. $Y = X * h$
13. $S_Y(f) = S_X(f)|H(f)|^2$
14. $R_Y(\tau) = R_X(\tau) * h(\tau) * h(-\tau)$
15. $\sigma^2 = 4kTRW$, $S_N(f) = N_0/2$, $P_Y = N_0W$ (8)